

AI-Driven Power Optimisation Of High-Performance Networking Chips In N3E Node



AI is revolutionising power management in power-hungry networking chips, optimising efficiency and performance in real time. Here is how it is reshaping TSMC's N3E process node.



The author, **VINAYAK RAMACHANDRA ADKOLI**, is BE in Industrial Production and has served as a lecturer in three different polytechnics for ten years. He is also a freelance writer and cartoonist.

With high-speed data processing and network connectivity becoming more demanding, networking chips must balance peak performance and power efficiency. TSMC's N3E process node, an advanced evolution of its 3nm technology, presents new opportunities for power optimisation in high-performance networking chips. Yet, this innovation also brings fresh design challenges, such as power leakage, thermal hotspots, and dynamic workloads.

Artificial intelligence (AI) is proving to be a game-changer in overcoming these hurdles. Techniques like machine learning (ML) and predictive analytics allow real-time power optimisation, ensuring chips operate with greater energy efficiency without compromising performance. By integrating AI-driven approaches, power management in networking chips built on the N3E node is reaching new levels of sophistication.

The N3E node

The N3E process node by TSMC delivers significant power and performance advantages

over previous technologies.

Key benefits include:

- 15–20% power savings at the same performance levels
 - 18% performance improvement at equivalent power
 - Higher transistor density while maintaining process reliability
- Despite these advantages, advanced nodes like N3E face new challenges:
- Dynamic power dissipation caused by the rapid switching of transistors in high-bandwidth chips
 - Increased leakage currents as transistors shrink, leading to idle power losses
 - Thermal management issues due to dense transistor placement and heavy workloads

High-performance networking chips, which manage data routing, packet processing, and real-time communications, are particularly vulnerable to these issues. Traditional power optimisation techniques are no longer sufficient, making AI-driven solutions essential for energy-efficient chip performance.